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WORK INFORMATION PROJECTION SYSTEM AND RELATIVE INFORMATION CALIBRATION METHOD

Abstract

A work information projection system (1) includes a projector (12), a camera (14), and a controller. The projector (12) projects an auxiliary image (101) onto a workplace. The camera (14) captures the workplace. During a work, the controller controls the projector (12) based on a position of the projector (12) to project the auxiliary image onto the workplace. During a calibration process of calibrating relative information, the camera (14) captures a calibration image projected by the projector (12) onto a calibration jig (32) through a reflector (15). The controller performs the calibration process of calibrating the relative information based on the calibration image captured by the camera (14).

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Background/Summary

TECHNICAL FIELD

[0001] The present invention mainly relates to a technology for calibrating a positional relation between a camera and a projector in a work information projection system that projects an auxiliary image for assisting a work onto a workplace.

BACKGROUND ART

[0002] PTL 1 discloses an information presentation apparatus for work assistance that includes a projector, a camera, a display, and a processing controller. The projector projects a projection target onto a workplace. The camera captures a plane of projection of the projector. The display displays the image captured by the camera. When a worker selects a projection target on the display using a mouse or a keyboard, the processing controller performs a process such as hiding the projection target selected by the worker. In addition, PTL 1 discloses a process of adjusting a positional relation between the projector and the camera using the camera to capture an image projected by the projector.

PRIOR-ART DOCUMENTS

Patent Documents

[0003] PTL 1: Japanese Patent Application Publication No. 2005-322055.

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

[0004] In PTL 1, the camera is arranged at a place where it can capture an image projected by the projector. A position onto which the projector projects the image is, in other words, a position where the work is performed. Thus, at this position, the position and shape of objects change as the work progresses. Therefore, when estimating the position of the camera based on the image captured by the camera, changes in the position and shape of object is a disturbance. As a result, the accuracy of estimation of the position of the camera may decrease. However, as disclosed in PTL 1, in order to calibrate the positional relation between the camera and the projector, it is necessary that the camera captures the image projected by the projector.

[0005] The present invention is made in view of the situation described above, and its main purpose is to provide a system that appropriately calibrates the relative positions of a camera and a projector during a calibration process, wherein a disturbance in an image captured by the camera during a work is less likely to occur.

Means for Solving the Problems

[0006] The problem to be solved by the present invention is as described above. The means to solve this problem and the effects thereof will be described below.

[0007] A first aspect of the present invention provides a work information projection system with a configuration described below. That is, the work information projection system includes a projector, a camera, and a controller. The projector projects an auxiliary image that assists a work onto a workplace. The camera captures the workplace. The controller creates the auxiliary image and transmits it to the projector. During a calibration process of calibrating relative information that indicates a relative positional relation between the camera and the projector, the camera captures a calibration image projected by the projector onto a calibration jig through a reflector. The controller performs the calibration process of calibrating the relative information based on the calibration image captured by the camera.

[0008] A second aspect of the present invention provides a work information projection method as

described below. That is, in the work image projection method, a position is estimated based on an image captured by a camera and an auxiliary image that assists a work is projected onto a workplace using a projector attached to move integrally with the camera. The work image projection method includes a calibration process of calibrating relative information that indicates a positional relation between the camera and the projector. In the calibration process, the projector projects a calibration image onto a calibration jig. In the calibration process, the camera captures the calibration image through a reflector. In the calibration process, the relative information is calibrated based on the calibration image captured by the camera.

[0009] By performing the calibration process using the reflector, it is possible to arrange the camera to face in a direction significantly different from a direction that the projector faces in. Therefore, for example, by capturing a position where no work is being performed, the accuracy of the detection of the position of the camera becomes high since a disturbance is less likely to occur in the image captured by the camera. As a result, the projector can project the auxiliary image at an accurate position.

Effects of the Invention

[0010] The present invention can provide a system that appropriately calibrates relative positions of a camera and a projector during a calibration process, wherein a disturbance in an image captured by the camera during a work is less likely to occur.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a schematic view of a work information projection system according to one embodiment of the present invention.

[0012] FIG. 2 is a block diagram of a work information projection system.

[0013] FIG. 3 is a flowchart of a calibration process.

[0014] FIG. 4 is a side view illustrating a calibration process being performed using a reflector.

[0015] FIG. 5 is a flowchart of processes performed during a work.

[0016] FIG. 6 is a schematic drawing of a work information projection system according to a first modification.

[0017] FIG. 7 is a drawing comparing configuration of a work information projection system according to a second modification during a calibration process and during a work.

[0018] FIG. 8 is a flowchart during a work according to a second modification.

EMBODIMENT FOR CARRYING OUT THE INVENTION

[0019] An embodiment of the present invention will be described below with reference to

[0020] the drawings. First, with reference to FIG. 1 and FIG. 2, outlines of a work information projection system 1 and a work information projection method will be described below.

[0021] The work information projection system 1 of the present embodiment is arranged in a workplace. The workplace is a place where a work is performed, such as a factory, an office, or an institution. The work is something that a worker does to an object by hand, by using a tool, or by operating a machine, such as assembling parts, painting, cleaning, or transfer. In the present embodiment, the worker performs the work of attaching a part to a workpiece 31 arranged in a factory.

[0022] The work information projection system 1 projects an auxiliary image 101 onto the workplace. The auxiliary image 101 is an image that assists the work, such as an image that conveys content of the work, a position to perform the work, an order of the work, or the like to the worker. As shown in FIG. 1, the auxiliary image 101 of the present embodiment is projected onto the workpiece 31 and indicates names of parts to be attached and positions where to attach them. The work information projection system 1 recognizes the position of the workpiece 31 (details of

this process will be described below) and projects the auxiliary image **101** onto the workpiece **31** at an appropriate position.

[0023] As shown in FIG. **1** and FIG. **2**, the work information projection system **1** includes a cart **11**, a projector **12**, a camera mount **13**, a camera **14**, and a controller **20**. In the following description, unless otherwise specified, the meaning of the term “position” shall include not only a location of an object, but also a direction in which the object is facing. Thus, for example, the meaning of the term “positional relation” includes not only the relative positions of two objects, but also their relative orientations.

[0024] The cart **11** includes wheels and a base. The cart **11** of the present embodiment does not include a drive source and moves in the workplace by being pushed by the worker. The cart **11** may include a drive source and be capable of autonomous travel. The projector **12**, the camera mount **13**, the camera **14**, and the like are placed on the base. The cart **11** may be movable along a rail arranged on the floor, ceiling, or the like. The cart **11** is not an essential component and may be omitted.

[0025] The projector **12** is placed on the cart **11**. The projector **12** projects the auxiliary image **101** under the control by the controller **20**.

[0026] The camera mount **13** is arranged on the top face of the projector **12**. The camera **14** can be attached to the camera mount **13**. This prevents changes in the relative positions of the projector **12** and the camera **14**. In other words, the projector **12** and the camera **14** move integrally. The arrangement method of the projector **12** and the camera **14** may be different from the present embodiment as long as the relative positions of the projector **12** and the camera **14** do not change. For example, a projector mount and a camera mount may be formed on the cart **11**. Alternatively, a projector mount and a camera mount may be formed on a supporting member attached to the cart **11**.

[0027] The camera **14** is a stereo camera that includes two imaging elements and they capture the workplace individually. The two imaging elements are arranged in an appropriate distance. These imaging elements are, for example, CCDs (Charge Coupled Devices). The two imaging elements operate in synchronization with each other and create a pair of image data by capturing the workplace at the same time. In the present embodiment, since it is supposed that information detected in real time is projected as an auxiliary image, it is preferable that the camera **14** captures an image for more than one time per, for example, second.

[0028] Furthermore, the camera **14** includes an image processor that processes this pair of image data. The image processor performs a known stereo matching process on the pair of image data obtained by the camera **14** to determine a distance between the positions that the two images correspond to (a disparity). The disparity increases in inverse proportion to the distance to the object captured in the image as the distance becomes shorter. The image processor creates a range image in which each pixel of the image data is linked to information about the distance based on this disparity.

[0029] The camera **14** has a configuration in which two imaging elements are arranged within one housing. Alternatively, two separate cameras may be combined to form the camera **14**. Further, the image processor may be arranged in an apparatus other than the camera **14** (for example, the controller **20** as described below).

[0030] The controller **20** is a computer that includes a CPU, a ROM, a RAM, and the like. The controller **20** of the present embodiment is arranged at the cart **11**. The controller **20** can communicate with the projector **12** and the camera **14** via signal lines that are not shown in the drawings. The controller **20** may be arranged outside the cart **11**. In this case, the controller **20** communicates with the projector **12** and the camera **14** wirelessly, for example.

[0031] The controller **20** creates the auxiliary image **101** based on the range image received from the camera **14** and other information and transmits the auxiliary image **101** to the projector **12**. As shown in FIG. **1**, the controller **20** includes a communicator **21**, an analyzer **22**, an image creator

23, and a projection controller **24**. Each part that the controller **20** includes is a conceptually divided part of the controller **20** for each process performed by the controller **20** (for each function the controller **20** has). Although the controller **20** of the present embodiment is realized by one computer, the controller **20** may be configured with more than one computer. In this case, these computers are connected via a network.

[0032] The communicator **21** is a communication module for communicating with the projector **12** and the camera **14** and includes, for example, a connector for connecting the signal line, an antenna for wireless communication, or the like. The communicator **21** receives the image captured by the camera **14** and transmits the auxiliary image **101** created by the image creator **23** to the projector **12**.

[0033] The analyzer **22** performs a SLAM (Simultaneous Localization and Mapping) process on the range image captured by the camera **14**. The analyzer **22** analyzes the range image to create map information (environmental map) that indicates shapes and positions of objects in the workplace and to estimate the position of the camera **14**. The objects in the workplace are, for example, a facility, an apparatus, a tool, or the workpiece **31** arranged in the workplace.

[0034] Since the SLAM process is well known, it will be briefly described below. That is, the analyzer **22** analyzes the range image, sets an appropriate feature point, and acquires the movement thereof. Then, the analyzer **22** extracts and tracks more than one feature point in the range image to obtain data that expresses the movements of the feature points in a plane that corresponds to the image using vectors. The analyzer **22** creates map information based on this data. As described above, the map information is data that indicates shapes and positions of objects in the workplace, and more specifically, it is data that indicates three-dimensional positions of the extracted feature points (point cloud). Based on a change in the positions and the distances of the input feature points and the positions of the feature points in the map information, the analyzer **22** estimates a change in the position of the camera **14**. The camera **14** may be a monocular camera that includes a single imaging element. The SLAM process as described above may also be performed using an image captured by the monocular camera.

[0035] The image creator **23** creates the auxiliary image **101**. The controller **20** stores work information which is information about the work. The work information of the present embodiment is the names of the parts to be attached to the workpiece **31** and the positions where to attach the parts. The image creator **23** creates the auxiliary image **101** to be projected by the projector **12** based on the work information and the position of the projector **12**. The position of the projector **12** is estimated based on the position of the camera **14** estimated by the analyzer **22** and the relative information that indicates the positional relation between the projector **12** and the camera **14**. Since the projector **12** and the camera **14** are so fixed that they do not move relative to each other, the relative information do not change.

[0036] The projection controller **24** transmits the auxiliary image **101** created by the image creator **23** to the projector **12** to project the auxiliary image **101**. With these configurations described above, the auxiliary image **101** can be projected onto the workplace.

[0037] Then, the calibration process of calibrating the relative information will be described below with reference to FIG. 3 and FIG. 4.

[0038] First, the camera **14** is attached to the camera mount **13**. This prevents the positional relation between the projector **12** and the camera **14** from changing. When the camera **14** is attached to the camera mount **13**, the direction of an optical axis **103** of the camera **14** is significantly different from the direction of an optical axis (projection axis) **104** of the projector **12**. The optical axis is a straight line extending in an axial direction from a point that passes an optical element (an imaging element, a light emitting element). In the present embodiment, the optical axis **103** of the camera **14** passes outside the range in which the projector **12** can project projection light. The optical axis **104** of the projector **12** is parallel to a horizontal direction, while the optical axis **103** of the camera **14** extends upward with respect to a horizontal plane. The horizontal plane is a plane perpendicular

to a vertical direction. The optical axis **103** of the camera **14** may extend downward with respect to a horizontal plane. The difference in the elevation angles of the optical axis **103** of the camera **14** and the optical axis **104** of the projector **12** is, for example, 20 degrees or more, 30 degrees or more, or 45 degrees or more.

[0039] Since the direction of the optical axis **103** of the camera **14** is significantly different from the direction of the optical axis **104** of the projector **12**, the camera **14** cannot directly capture the image projected by the projector **12**. Therefore, in the present embodiment, a reflector **15** for the calibration is arranged to cover the optical axis **103** of the camera **14**. The reflector **15** is a member that specularly reflects visible light, specifically, it is a mirror or a prism. Arranging the reflector **15** allows the camera **14** to capture the image projected by the projector **12** through the reflector **15**. The reflector **15** can be attached to or detached from a component (for example, the cart **11**) of the work information projection system **1**. In the present embodiment, the reflector **15** is arranged only during the calibration process and is removed during the work of projecting the auxiliary image **101** onto the workplace. A marker **16** is attached to the reflector **15**.

[0040] Then, a calibration jig **32** for the calibration process is arranged in front of the projector **12**. This completes the preparation for the calibration process.

[0041] Upon receipt of an instruction to start the calibration process, the controller **20** transmits a calibration image **102** to the projector **12**. The projector **12** projects the transmitted calibration image **102** onto the calibration jig **32** (S101, first process). Then, through the reflector **15**, the camera **14** captures the calibration image **102** projected onto the calibration jig **32** (S102, second process). The image captured by the camera **14** also includes the marker **16**.

[0042] Then, based on the position, the size, the degree of distortion, or the like of the marker **16** included in the image captured by the camera **14**, the controller **20** performs a known analysis process to calculate the relative position (that is, positional relation between the camera **14** and the reflector **15**) of the marker **16** with respect to the camera **14** (S103).

[0043] Then, based on the calibration image captured by the camera **14**, the controller **20** calculates the positional relation between a virtual camera **14a** and the projector **12** (S104). The position of the virtual camera **14a** can be calculated based on the positional relation between the camera **14** and the reflector **15**. Specifically, the virtual camera **14a** is at a position that is plane-symmetrical to the camera **14** with the reflection plane of the reflector **15** as a symmetry plane. To calculate the positional relation between the virtual camera **14a** and the projector **12** based on the calibration image, a conventional calibration process can be used. In other words, based on the position, the size, the degree of distortion, or the like of a displayed object included in the calibration image, the positional relation between the virtual camera **14a** and the projector **12** can be calculated.

[0044] Then, the controller **20** calculates the relative information which is the positional relation between the camera **14** and the projector **12** (S105, third process). In the step S104, the positional relation between the virtual camera **14a** and the projector **12** has already been calculated.

Furthermore, as described above, the positional relation between the camera **14** and the virtual camera **14a** has already been calculated. Therefore, based on these two positional relations, the relative information can be calculated. This completes the calibration process.

[0045] As described above, even when the camera **14** cannot directly capture the calibration image projected by the projector **12**, the calibration of the relative information which is the positional relation between the camera **14** and the projector **12** can be performed by using the reflector **15**.

[0046] Next, the process performed during the work when the auxiliary image **101** is projected onto the workplace will be described below with reference to FIG. 1 and FIG. 5.

[0047] First, the issues exist in the conventional technology (PTL 1) will be described. In the conventional technology, the camera and the projector face in almost the same direction. Therefore, the camera captures the work position where the worker performs the work. However, at the work position, positions, sizes, shapes, or the like of objects change as the work progresses, so that the feature points for the SLAM process may change (that is, they are disturbances) and the accuracy

of the estimation of the position of the camera may become low. As a result, the accuracy of the position at which the auxiliary image is projected by the projector becomes low. Furthermore, since the camera captures the auxiliary image projected by the projector, the auxiliary image may be determined as a feature point (that is, it may become a disturbance). If the auxiliary image is determined as a feature point, errors will be accumulated because the position at which the auxiliary image is projected is adjusted according to an error in the estimated position of the camera and the projected auxiliary image is captured by the camera and processed as a feature point. As a result, the accuracy of the position at which the auxiliary image is projected by the projector becomes significantly low.

[0048] When the size of the workpiece is large and the surface of the workpiece has few undulations (for example, the workpiece **31** shown in FIG. **1**), the number of the feature points is so small that the accuracy of the estimation of the position of the camera becomes low even if the camera is arranged close to the workpiece. In this case, with the conventional technology, the distance between the camera and the workpiece should be increased. However, if the distance between the camera and the workpiece is increased, the required output of the projector increases so that the cost increases.

[0049] In contrast, in the present embodiment, the accuracy of the position at which the auxiliary image **101** is projected by the projector **12** is higher and the position of the camera **14** is less restricted than in the conventional technology. The specifics thereof are explained below.

[0050] First, the analyzer **22** of the controller **20** performs the processes described above on the image captured by the camera **14** to estimate the positions of the origin and the camera **14** in the workplace (**S201**). The position of the camera **14** in the workplace is the position of the camera **14** in the coordinate system of the environmental map created with respect to the workplace (the same applies to the projector **12** and the workpiece **31**). In the following description, the position of the camera **14** or the like in the workplace may be referred to simply as the position of the camera **14**.

[0051] Since the direction that the camera **14** of the present embodiment faces in is significantly different from the direction that the projector **12** faces in, the camera **14** does not capture the auxiliary image **101**. The camera **14** captures a ceiling member **33** which is, for example, a beam or a lighting apparatus arranged on the ceiling. Since a change of, for example, the position of an object with respect to the ceiling member **33** is less likely to occur in contrast to the work position, the accuracy of the estimation of the self-position of the camera **14** is higher than the conventional technology. Even if the camera **14** is arranged near the workpiece **31**, the camera **14** captures the ceiling member **33**, not the workpiece **31**. Therefore, the position of the camera **14** is less restricted. The camera **14** may capture a wall, a shelf, a machine, or the like instead of or in addition to the ceiling member **33**.

[0052] Then, the analyzer **22** of the controller **20** estimates the position of the projector **12** in the workplace based on the position of the camera **14** in the workplace estimated in step **S201** and the relative information obtained by the calibration process (**S202**).

[0053] Then, the image creator **23** of the controller **20** creates the auxiliary image **101** based on the position of the projector **12** with respect to the position of the workpiece **31** in the workplace (**S203**). Since the positional relation between the workpiece **31** and the projector **12** is known, how an image is displayed on the workpiece **31** when the image is projected by the projector **12** can be obtained by calculation. Therefore, the image creator **23** can create the auxiliary image **101** that can be projected onto a predetermined position on the workpiece **31** (position where a part should be attached) at a predetermined size (the same size as the actual size of the part).

[0054] Then, the projection controller **24** of the controller **20** transmits the auxiliary image **101** created in step **S203** to the projector **12** and instructs it to project the auxiliary image **101** (**S204**).

[0055] The controller **20** repeats the process shown in FIG. **5**. Therefore, even when the cart **11** moves, the auxiliary image **101** projected onto the workpiece **31** does not change because the auxiliary image **101** is created with the size and the position corresponding to the position after the

movement.

[0056] When the work shown by the auxiliary image **101** is completed, the worker reports it to the controller **20** using an information terminal or the like. In response, the controller **20** creates the auxiliary image **101** for the next work and projects it. In addition to the camera **14**, a monocular camera or a camera may be arranged to capture the work position and the controller **20** may determine whether the work is completed based on the image captured by them.

[0057] Next, a first modification of the above embodiment will be described with reference to FIG. **6**. FIG. **6** is a schematic drawing of a work information projection system **1** according to the first modification. In the descriptions of the first modification and subsequent modifications, the components identical or similar to those of the above-described embodiment may be marked with the same reference signs in the drawings and the description thereof may be omitted.

[0058] In the above-described embodiment, when the direction in which the projector **12** projects the auxiliary image **101** is defined as forward, the optical axis of the camera **14** includes a forward direction as a component. In contrast, in the first modification, the optical axis of the camera **14** does not include a forward direction as a component (it includes a backward direction as a component). This allows high accuracy of the estimation of the position of the camera **14** even in an environment where few feature points exist in front and above (for example, an environment where an only flat wall is arranged).

[0059] Next, a second modification will be described with reference to FIG. **7** and FIG. **8**. FIG. **7** is a drawing that compares the configuration of a work information projection system **1** according to the second modification during a calibration process and during a work. FIG. **8** is a flowchart during the work according to the second modification.

[0060] In the above-described embodiment, the reflector **15** is used during the calibration process and the reflector **15** is removed when performing the work. In contrast, in the second modification, the reflector **15** is used not only during the calibration process but also during the work. Specifically, as shown in FIG. **7**, the work information projection system **1** includes a frame **17**, a mount **18**, a motor (driver) **41**, and an encoder (detector) **42**.

[0061] The frame **17** is a gantry-shaped (substantially U-shaped) frame and is shaped so as not to interfere with the camera **14** capturing image. The mount **18** is attached to the frame **17** in a way that it can relatively rotate. The reflector **15** is fixed to the mount **18**. In other words, the mount **18** and the reflector **15** integrally rotate.

[0062] The motor **41** is a motor wherein the amount of the rotation of the output shaft is controllable (for example, a stepping motor) and is controlled by the controller **20**. The motor **41** generates power to rotate the reflector **15**. The power generated by the motor **41** is transmitted to the mount **18** via a transmission mechanism not shown in the drawings. With this configuration, the direction that the reflector **15** faces in can be changed. Although the rotational axis of the reflector **15** is parallel to a horizontal direction in the present embodiment, it may be not parallel to it. The encoder **42** is attached to the mount **18** and detects the rotation angle of the mount **18** (that is, the direction that the reflector **15** faces in).

[0063] With the above configuration, the direction that the reflector **15** faces in can be changed according to the amount of the rotation instructed by the controller **20**. Since the reflector **15** is arranged to cover the optical axis **103** of the camera **14**, the position to be captured by the camera **14** can be changed by changing the direction that the reflector **15** faces in.

[0064] A drawing described as “During Calibration Process” in FIG. **7** shows the direction that the reflector **15** faces in during the calibration process (a first direction). During the calibration process, the direction in which the reflector **15** faces is so controlled that the camera **14** can capture the calibration image **102**.

[0065] A drawing described as “During Work” in FIG. **7** shows the direction that the reflector **15** faces in during the work (a second direction). As described above, when the camera **14** captures the auxiliary **101** during the work, the auxiliary image **101** becomes a disturbance. Therefore, during

the work, the direction in which the reflector **15** faces is so controlled that the camera **14** captures the position that does not include the auxiliary image **101**.

[0066] In the configuration of the second modification, there is no need to attach and remove the reflector **15** during the calibration process and the work. Therefore, the time and effort required for switching between the calibration process and the work can be reduced.

[0067] The controller **20** of the second modification performs the process shown in FIG. **8** during the work in order to maintain the accuracy of the estimation of the position of the camera **14** in the workplace high.

[0068] The controller **20** calculates the number of feature points included in the image captured by the camera **14** (S301). Then, the controller **20** determines whether the number of the feature points calculated in step S301 is equal to or exceeds a threshold (S302). In a case where the number of the feature points is too small, the position of the camera **14** may not be accurately estimated.

[0069] If the controller **20** determines that the number of the feature points is less than the threshold, it changes the direction in which the reflector **15** faces by controlling the motor **41** (S303). For example, to change the direction that the reflector **15** faces in by N degrees, the controller **20** rotates the output shaft of the motor **41** until the value detected by the encoder **42** changes by N degrees.

[0070] Changing the direction the reflector **15** faces in changes the position that the camera **14** captures. Then, the controller **20** again performs the processes in step S301 and S302. If the number of the feature points included in the image captured by the camera **14** becomes equal to or exceeds the threshold, the controller **20** does not perform the process of changing the direction the reflector **15** faces in. In other words, the controller **20** continues to change the direction the reflector **15** faces in until the number of the feature points included in the image captured by the camera **14** becomes equal to or exceeds the threshold.

[0071] In this manner, even when the camera captures the position with few feature points (for example, a flat wall), changing the direction the reflector **15** faces in allows the camera **14** to capture another position with many feature points. Therefore, this maintains high accuracy of the estimation of the position of the camera **14**. In addition, it is not preferable that the camera **14** captures the auxiliary image **101**, the cart **11**, or the like during the work. Therefore, it is preferable to set a maximum limit and a minimum limit for the range of changing the direction that the reflector **15** faces in.

[0072] As the direction that the reflector **15** faces in changes, the image captured by the camera **14** changes. Therefore, in order to update the map information that has already been created, it is necessary to identify which direction in the coordinate system of the environmental map the image captured by the camera **14** through the reflector **15** was captured from. In this regard, in the present embodiment, the accurate direction that the reflector **15** faces in is detected by the encoder **42**.

Therefore, even if the direction that the reflector **15** face in is changed, the position of the camera **14** can be estimated and the existing map information can be updated by using the amount of change in the direction that the reflector **15** faces in (a value that indicates how much it changed from a reference direction) and the existing map information.

[0073] As described above, the work information projection system **1** of the above-described embodiment includes the projector **12**, the camera **14**, and the controller **20**. The projector **12** projects the auxiliary image **101** that assists the work onto the workplace. The camera **14** captures the workplace. The controller **20** creates the auxiliary image **101** and transmits it to the projector **12**. During the calibration process of calibrating the relative information that indicates the relative positional relation between the camera **14** and the projector **12**, the camera **14** captures the calibration image **102** projected by the projector **12** onto the calibration jig **32** through the reflector **15**. The controller **20** performs the calibration process of calibrating the relative information based on the calibration image **102** captured by the camera **14**.

[0074] Performing the calibration process using the reflector **15** allows the camera **14** to be pointed

in a direction significantly different from the direction that the projector **12** faces in. Therefore, capturing a position where no work is being performed, for example, makes a disturbance less likely to occur in the image captured by the camera **14** so that the accuracy of the detection of the position of the camera **14** becomes high. As a result, the projector **12** can project the auxiliary image at an accurate position.

[0075] In the work information projection system **1** of the above-described embodiment, during the work when the auxiliary image **101** is projected onto the workplace, the controller **20** estimates the position of the camera **14** based on the image captured by the camera **14**. The controller **20** estimates the position of the projector **12** based on the relative information and the position of the camera **14**. The controller **20** controls the projector **12** based on the position of the projector **12** to project the auxiliary image **101** onto the workplace.

[0076] This allows the auxiliary image to be projected at an accurate position during the work.

[0077] In the work information projection system **1** of the above-described embodiment, the camera **14** is so arranged that it moves integrally with the projector **12**.

[0078] This allows the accurate position of the projector **12** to be calculated and the auxiliary image because the positional relation between the camera **14** and the projector **12** does not change, and thus the auxiliary image can be projected at an accurate position.

[0079] In the work information projection system **1** of the above-described embodiment, during the work, the direction of the optical axis **103** of the camera **14** is different from the direction of the optical axis **104** of the projector **12**.

[0080] Furthermore, in the work information projection system **1** of the above-described embodiment, during the work, the optical axis **103** of the camera **14** is not included in the range in which the projector **12** can project the projection light.

[0081] This makes a disturbance even less likely to occur in the image captured by the camera **14** because the directions that the camera **14** faces in and the projector **12** faces in are significantly different.

[0082] In the work information projection system **1** of the above-described embodiment, the direction that the reflector **15** faces in is changeable.

[0083] This allows the direction that the reflector **15** faces in to be changed as needed.

[0084] In the work information projection system **1** of the above-described embodiment, the direction that the reflector **15** faces in is changeable between the first direction with which the camera **14** can capture the calibration image **102** projected by the projector **12** and the second direction with which the camera **14** cannot capture the auxiliary image **101** projected by the projector **12**.

[0085] This allows the camera **14** to capture the calibration image **102** projected by the projector **12** during the calibration process. This also makes a disturbance even less likely to occur because the camera **14** does not capture the auxiliary image **101** projected by the projector **12** during the work.

[0086] The work information projection system **1** of the above-described embodiment includes the motor **41** that generates power to change the direction that the reflector **15** faces in. During the work, the camera **14** captures an image through the reflector **15**. The controller **20** changes the direction that the reflector **15** faces in by controlling the motor **41** when it determines that the number of the feature points included in the image captured by the camera **14** becomes less than the threshold.

[0087] This allows the camera **14** to capture the position that includes many feature points, so that the position of the camera **14** can be estimated even more accurately.

[0088] In the work information projection system **1** of the above-described embodiment, the camera **14** captures the calibration image **102** through the reflector **15** during the calibration process. The camera **14** captures the workplace without the reflector **15** during the work.

[0089] This allows a configuration in which the camera **14** captures the image projected by the projector **12** during the calibration process, meanwhile the camera **14** does not capture the image

projected by the projector **12** during the work.

[0090] In the work information projection system **1** of the above-described embodiment, during the calibration process, the controller **20** calculates the positional relation between the camera **14** and the reflector **15** based on the reflector **15** included in the image captured by the camera **14**. The controller **20** calculates the positional relation between the camera **14** and the virtual camera **14a** based on the positional relation between the camera **14** and the reflector **15**. The controller **20** calculates the positional relation between the projector **12** and the virtual camera **14a** based on the calibration image **102** reflected in the reflector **15** and calibrates the relative information.

[0091] This allows the appropriate calibration of the relative information.

[0092] In the work information projection system **1** of the above-described embodiment, the optical axis **103** of the camera **14** is upward with respect to a plane perpendicular to a vertical direction.

[0093] This allows the upper position where a disturbance is relatively less likely to occur to be captured. The optical axis of the camera **14** may also be downward with respect to a plane perpendicular to a vertical direction.

[0094] Although the preferred embodiment and the modifications of the present invention has been described above, the above-described configurations can be modified, for example, as follows.

[0095] The flowcharts shown in the above-described embodiment are disclosed just as examples and some processes may be omitted, the contents of some processes may be changed, or new processes may be added. For example, in the flowchart shown in FIG. **3**, the positional relation between the camera **14** and the reflector **15** may be firstly calculated and the calibration may be projected and captured after that (steps **S101** and **S102** may be performed after step **S103**).

[0096] In the above-described embodiment, the projector **12** and the camera **14** are movable by the cart **11**. Alternatively, the projector **12** and the camera **14** may be configured to move together with an operator, for example, being held in the operator's hand or attached to the operator.

[0097] In the above-described embodiment, the auxiliary image **101** is projected onto the workpiece **31**, but the auxiliary image may be projected onto an object other than the workpiece **31** (for example, a wall, a tool, a machine, or the like).

DESCRIPTION OF THE REFERENCE SIGNS

[0098] **1** Work information projection system

[0099] **11** Cart

[0100] **12** Projector

[0101] **14** Camera

[0102] **15** Reflector

[0103] **16** Marker

[0104] **20** Controller

Claims

1. A work information projection system, comprising: a projector that projects an auxiliary image that assists work onto a workplace; a camera that captures the workplace; a memory; and a controller, connected to the memory, that creates the auxiliary image and transmits it to the projector, wherein, during a calibration process of calibrating relative information that indicates a relative positional relation between the camera and the projector: the camera captures a calibration image projected by the projector onto a calibration jig through a reflector, and the controller performs the calibration process of calibrating the relative information based on the calibration image captured by the camera.

2. The work information projection system according to claim 1, wherein, during the work when the auxiliary image is projected onto the workplace: the controller estimates a position of the camera based on an image captured by the camera, the controller estimates a position of the projector based on the relative information and the position of the camera, and the controller

controls the projector based on the position of the projector to project the auxiliary image onto the workplace.

3. The work information projection system according to claim 1, wherein the camera moves integrally with the projector.

4. The work information projection system according to claim 1, during the work when the auxiliary image is projected onto the workplace, a direction of an optical axis of the camera is different from a direction of an optical axis of the projector.

5. The work information projection system according to claim 4, wherein during the work, the optical axis of the camera is not included in a range in which the projector can project projection light.

6. The work information projection system according to claim 1, wherein a direction that the reflector faces in is changeable.

7. The work information projection system according to claim 6, wherein; the direction that the reflector faces in is changeable between a first direction with which the camera can capture the calibration image projected by the projector and a second direction with which the camera cannot capture the auxiliary image projected by the projector.

8. The work information projection system according to claim 7, further comprising: a driver that generates power to change the direction that the reflector faces in, wherein, during the work when the auxiliary image is projected onto the workplace: the camera captures the workplace through the reflector, and the controller changes the direction that the reflector faces in by controlling the driver when it determines that a number of feature points included in an image captured by the camera becomes less than a threshold.

9. The work information projection system according to claim 1, wherein; the camera captures the calibration image through the reflector during the calibration process, and the camera captures the workplace without the reflector during the work when the auxiliary image is projected onto the workplace.

10. The work information projection system according to claim 1, wherein, during the calibration process: the controller calculates a positional relation between the camera and the reflector based on the reflector included in an image captured by the camera, the controller calculates a positional relation between the camera and a virtual camera based on the positional relation between the camera and the reflector, and the controller calculates a positional relation between the projector and the virtual camera based on the calibration image reflected in the reflector and calibrates the relative information.

11. The work information projection system according to claim 1, wherein an optical axis of the camera is upward or downward with respect to a plane perpendicular to a vertical direction.

12. A work information projection method, comprising: capturing an image by a camera; calibrating relative information that indicates a positional relation between the camera and a projector; and projecting an auxiliary image onto a workplace using the projector that moves integrally with the camera, wherein the calibrating includes: projecting a calibration image onto a calibration jig by the projector; and capturing the calibration image through a reflector by the camera; and a process of calibrating the relative information based on the calibration image captured by the camera.

13. The work information projection method according to claim 12, wherein; the reflector is detachable, the calibrating is performed with the reflector attached, and the projecting the auxiliary image onto the workplace is performed with the reflector detached.
